

Safety data sheet

Regulation (EC) 1907/2006 and 1272/2008

Printing date: 18.08.2023

Version number: 1

Revision date: 17.08.2023

SECTION 1: Identification of the substance/mixture and of the company/undertaking

- **1.1 Product identifier**
- **Trade name:** REDUCED IRON POWDER SERIES
- **UFI:** VD80-X0AD-V00Y-V9YD
- **1.2 Relevant identified uses of the substance or mixture and uses advised against**
- **Application of the substance / the mixture:** POWDER METALLURGY PARTS
- **1.3 Details of the supplier of the safety data sheet**
- **Manufacturer / Supplier:** SHANDONG LUYIN NEW MATERIAL TECHNOLOGY CO.,LTD
- **Full address:** JIULONG AVENUE, GANGCHENG, LAIWU CITY, SHANGDONG, CHINA
- **Phone number:** 13906349681
- **Email:** 13906349681@126.com
- **Only Representative / other EU contact point:** Not available
- **1.4 Emergency telephone number:**
IRELAND
National Poisons Information Centre
Tel: +353 (01) 809 2566 (For healthcare professionals)
+353 (01) 809 2166 (For public; 8am - 10pm)
- **1.5 Reference Number:** CANEC23008212201,GZP23-010446

SECTION 2: Hazards identification

- **2.1 Classification of the substance or mixture**
- **Classification according to Regulation (EC) No 1272/2008**
- 
GHS02 flame

 Flam. Sol. 2 H228 Flammable solid.
 Self-heat. 2 H252 Self-heating in large quantities; may catch fire.
- **Information concerning particular hazards for human and environment:**
The product has to be labelled due to the calculation procedure of Regulation (EC) No.1272/2008.
- **Classification system:**
The classification is according to the latest edition of EU Regulation (EC) No. 1272/2008, and extended by company and literature data.
- **2.2 Label elements**
- **Labelling according to Regulation (EC) No. 1272/2008**
The product is classified and labelled according to the CLP regulation.
- **Hazard pictograms**
- 

GHS02
- **Signal word** Warning
- **Hazard-determining components of labelling:** Not applicable
- **Hazard statements**
H228 Flammable solid.
H252 Self-heating in large quantities; may catch fire.
- **Precautionary statements**
P210 Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
P235+P410 Keep cool. Protect from sunlight.
P240 Ground and bond container and receiving equipment.

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- P241 Use explosion-proof [electrical/ventilating/lighting] equipment.
 P280 Wear protective gloves/protective clothing/eye protection/face protection/hearing protection.
 P420 Store separately.

- **2.3 Other hazards:**
- **Results of PBT and vPvB assessment**
- **PBT:** Not applicable
- **vPvB:** Not applicable
- **Determination of endocrine-disrupting properties** Not applicable

SECTION 3: Composition/information on ingredients

- **3.2 Mixtures**
- **Description:**
 Mixture of the substances listed below with nonhazardous additions.
 For the wording of the listed hazard statements refer to section 16.

- **Composition:**

CAS: 7439-89-6 EINECS: 231-096-4	Iron ⚠ Flam. Sol. 2, H228; Self-heat. 2, H252	≥97%
CAS: 7782-44-7 EINECS: 231-956-9 Index number: 008-001-00-8	Oxygen ⚠ Ox. Gas 1, H270; ⚠ Press. Gas (Comp.), H280	≤0.8%
CAS: 7439-96-5 EINECS: 231-105-1	Manganese ⚠ Flam. Sol. 2, H228 Substance with a Union workplace exposure limit	≤0.7%
CAS: 7440-21-3 EINECS: 231-130-8	Silicon Substance with a Union workplace exposure limit	≤0.3%
CAS: 7440-44-0 EINECS: 231-153-3	Carbon Substance with a Union workplace exposure limit	≤0.1%
CAS: 7704-34-9 EINECS: 231-722-6 Index number: 016-094-00-1	Sulfur ⚠ Skin Irrit. 2, H315	≤0.06%
CAS: 7723-14-0 EINECS: 231-768-7 Index number: 015-002-00-7	Red phosphorus ⚠ Flam. Sol. 1, H228; Aquatic Chronic 3, H412 Substance with a Union workplace exposure limit	≤0.04%

SECTION 4: First aid measures

- **4.1 Description of first aid measures**
- **General description:** No special measures required.
- **After inhalation:** Supply fresh air; consult doctor in case of complaints.
- **After skin contact:** Generally the product does not irritate the skin.
- **After eye contact:** Rinse opened eye for several minutes under running water.
- **After swallowing:** If symptoms persist consult doctor.
- **4.2 Most important symptoms and effects, both acute and delayed:**
 No further relevant information available.
- **4.3 Indication of any immediate medical attention and special treatment needed:**
 No further relevant information available.

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SECTION 5: Firefighting measures

- **5.1 Extinguishing media**
- **Suitable extinguishing agents:** Use fire extinguishing methods suitable to surrounding conditions.
- **5.2 Special hazards arising from the substance or mixture:** No further relevant information available.
- **5.3 Advice for firefighters**
- **Protective equipment:** No special measures required.

SECTION 6: Accidental release measures

- **6.1 Personal precautions, protective equipment and emergency procedures:** Not required.
- **6.2 Environmental precautions:** Do not allow to enter sewers/ surface or ground water.
- **6.3 Methods and material for containment and cleaning up:** Pick up mechanically.
- **6.4 Reference to other sections:**
See Section 7 for information on safe handling.
See Section 8 for information on personal protection equipment.
See Section 13 for disposal information.

SECTION 7: Handling and storage

- **7.1 Precautions for safe handling:**
No special measures required.
For the general occupational hygienic measures refer to Section 8.
- **Information about fire - and explosion protection:** No special measures required.
- **7.2 Conditions for safe storage, including any incompatibilities**
- **Requirements to be met by storerooms and receptacles:** No special requirements.
- **Information about storage in one common storage facility:** Not required.
- **Further information about storage conditions:** None.
- **7.3 Specific end use(s):** No further relevant information available.

SECTION 8: Exposure controls/personal protection

- **8.1 Control parameters**

- **Ingredients with limit values that require monitoring at the workplace:**

CAS: 7439-96-5 Manganese ($\leq 0.7\%$)

IOELV (EU)	Long-term value: $0.2 * 0.05^{**}$ mg/m ³ as Mn; *inhalable, **respirable fraction
AGW (Germany)	Long-term value: 0.02A; 0.2E mg/m ³ 8(II);DFG,Y,10, 20
VLEP (France)	Long-term value: $0.05 * 0.20^{**}$ mg/m ³ *fraction alvéolaire **inhalable; en manganèse
OEL (Ireland)	Short-term value: $3 * \text{mg/m}^3$ Long-term value: $0.2(I) 0.05(R) 0.02(R)^*$ mg/m ³ IOLEV; *fume

CAS: 7440-21-3 Silicon ($\leq 0.3\%$)

VLEP (France)	Long-term value: 10 mg/m ³
OEL (Ireland)	Long-term value: $10 * 4^{**}$ mg/m ³ *total inhalable **respirable dust

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CAS: 7440-44-0 Carbon ($\leq 0.1\%$)

AGW (Germany)	Long-term value: $1.25 \cdot 10^{**} \text{ mg/m}^3$ 2(II); *alveolengängig**eintembar; AGS, DFG, Y
OEL (Ireland)	Long-term value: $10 \cdot 4^{**} \text{ mg/m}^3$ *total inhalable **respirable dust

CAS: 7723-14-0 Red phosphorus ($\leq 0.04\%$)

AGW (Germany)	Long-term value: 0.01 E mg/m^3 2(II); AGS, Y
VLEP (France)	Short-term value: 0.3 mg/m^3 Long-term value: 0.1 mg/m^3
OEL (Ireland)	Short-term value: 0.3 mg/m^3 Long-term value: 0.1 mg/m^3

Regulatory information

IOELV (EU): (EU) 2019/1831

AGW (Germany): TRGS 900

VLEP (France): ED 1487 05.2021

OEL (Ireland): 2021 CoP for the Safety, Health and Welfare at Work

• **DNELs:** Not available• **PNECs:** Not available**Ingredients with biological limit values:****CAS: 7439-96-5 Manganese**

BGW (Germany)	20 $\mu\text{g/l}$ Untersuchungsmaterial: Vollblut Probennahmezeitpunkt: bei Langzeitexposition: am Schichtende nach mehreren vorangegangenen Schichten, Expositionsende bzw. Schichtende Parameter: Mangan
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• **Additional information:** The lists valid during the making were used as basis.**8.2 Exposure controls**

Based on the composition shown in Section 3, the following measures are suggested for occupational safety measure.

• **Appropriate engineering controls:** See Section 7 for information about design of technical facilities.• **Individual protection measures, such as personal protective equipment**• **Respiratory protection:** Not required.• **Hand protection:**

Protective gloves

The glove material has to be impermeable and resistant to the product/ the substance/ the preparation.

Due to missing tests no recommendation to the glove material can be given for the product/ the preparation/ the chemical mixture.

Selection of the glove material on consideration of the penetration times, rates of diffusion and the degradation.

Material of gloves:

The selection of the suitable gloves does not only depend on the material, but also on further marks of quality and varies from manufacturer to manufacturer. As the product is a preparation of several substances, the resistance of the glove material can not be calculated in advance and has therefore to be checked prior to the application.

Penetration time of glove material:

The exact break through time has to be found out by the manufacturer of the protective gloves and has to be observed.

• **Eye/face protection:** Not required.

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- **Body protection:** Protective work clothing
- **Thermal hazards:** Not required for normal conditions of use.
- **Environmental exposure controls:**
Control measures must be made in accordance with Community environmental protection legislation.

SECTION 9: Physical and chemical properties

· 9.1 Information on basic physical and chemical properties

- **Physical state:** Powder
- **Colour:** Silver-grey
- **Odour:** Not
- **Odour threshold:** Not available
- **Melting point/Freezing point:** 1538 °C
- **Boiling point or initial boiling point and boiling range:** Not available
- **Flammability:** Flammable.
- **Lower and upper explosion limit**
- **Lower:** Not available
- **Upper:** Not available
- **Flash point:** Not available
- **Auto-ignition temperature:** Not available
- **Decomposition temperature:** Not available
- **pH:** Not available
- **Viscosity**
- **Kinematic viscosity:** Not available
- **Dynamic viscosity:** Not available
- **Solubility**
- **Water:** Soluble
- **Partition coefficient n-octanol/water (log value):** Not available
- **Vapour pressure at 1455 °C:** 0.01 hPa (0.1hPa, 1617 °C)
- **Density and/or relative density**
- **Density:** 7.86 g/cm³
- **Relative density:** Not available
- **Relative vapour density:** Not available
- **Particle characteristics:** Not available

· 9.2 Other information

- **Appearance**
- **Form:** Powder

· Information with regard to physical hazard classes

- **Explosives:** Not applicable
- **Flammable gases:** Not applicable
- **Aerosols:** Not applicable
- **Oxidising gases:** Not applicable
- **Gases under pressure:** Not applicable
- **Flammable liquids:** Not applicable
- **Flammable solids:**
Flammable solid.
- **Self-reactive substances and mixtures:** Not applicable
- **Pyrophoric liquids:** Not applicable
- **Pyrophoric solids:** Not applicable
- **Self-heating substances and mixtures:**
Self-heating in large quantities; may catch fire.
- **Substances and mixtures, which emit flammable gases in contact with water:** Not applicable
- **Oxidising liquids:** Not applicable

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|--|----------------|
| · Oxidising solids: | Not applicable |
| · Organic peroxides: | Not applicable |
| · Corrosive to metals: | Not applicable |
| · Desensitised explosives: | Not applicable |
| · Other safety characteristics: | Not available |

SECTION 10: Stability and reactivity

- **10.1 Reactivity:** No further relevant information available.
- **10.2 Chemical stability:** Stable in dry air, easy to oxidize in humid air
- **10.3 Possibility of hazardous reactions:** No dangerous reactions known.
- **10.4 Conditions to avoid:** No further relevant information available.
- **10.5 Incompatible materials:** No further relevant information available.
- **10.6 Hazardous decomposition products:** No dangerous decomposition products known.

SECTION 11: Toxicological information

- **11.1 Information on hazard classes as defined in Regulation (EC) No 1272/2008**
- **Acute toxicity** Based on available data, the classification criteria are not met.

- **LD/LC50 values relevant for classification:**

CAS: 7439-89-6 Iron

Oral LD50 30,000 mg/kg (rat)

CAS: 7439-96-5 Manganese

Oral LD50 9,000 mg/kg (rat)

- **Skin corrosion/irritation:** Based on available data, the classification criteria are not met.
- **Serious eye damage/irritation:** Based on available data, the classification criteria are not met.
- **Respiratory or skin sensitization:** Based on available data, the classification criteria are not met.
- **Germ cell mutagenicity:** Based on available data, the classification criteria are not met.
- **Carcinogenicity:** Based on available data, the classification criteria are not met.
- **Reproductive toxicity:** Based on available data, the classification criteria are not met.
- **STOT-single exposure:** Based on available data, the classification criteria are not met.
- **STOT-repeated exposure:** Based on available data, the classification criteria are not met.
- **Aspiration hazard:** Based on available data, the classification criteria are not met.

- **11.2 Information on other hazards**

- **Endocrine disrupting properties:**

None of the ingredients is listed.

- **Other information:** No further relevant information available.

SECTION 12: Ecological information

- **12.1 Toxicity**
- **Aquatic toxicity:** No further relevant information available.
- **12.2 Persistence and degradability:** No further relevant information available.
- **12.3 Bioaccumulative potential:** No further relevant information available.
- **12.4 Mobility in soil:** No further relevant information available.
- **12.5 Results of PBT and vPvB assessment**
- **PBT:** Not applicable

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- **vPvB:** Not applicable
- **12.6 Endocrine disrupting properties:**
The product does not contain substances with endocrine disrupting properties.
- **12.7 Other adverse effects:** No further relevant information available.

SECTION 13: Disposal considerations

- **13.1 Waste treatment methods**
- **Recommendation:** Smaller quantities can be disposed of with household waste.
- **Uncleaned packaging:**
- **Recommendation:** Disposal must be made according to official regulations.

SECTION 14: Transport information

- | | |
|--|---|
| <ul style="list-style-type: none"> · 14.1 UN number or ID number · ADR/RID/ADN, IMDG, IATA | <p style="text-align: center;">UN3189</p> |
| <ul style="list-style-type: none"> · 14.2 UN proper shipping name · ADR/RID/ADN, IMDG, IATA | <p style="text-align: center;">METAL POWDER, SELF-HEATING, N.O.S.</p> |
| <ul style="list-style-type: none"> · 14.3 Transport hazard class(es) · ADR/RID/ADN, IMDG, IATA | <div style="text-align: center;">  </div> |
| <ul style="list-style-type: none"> · Class · Label | <p style="text-align: center;">4.2 Substances liable to spontaneous combustion.
4.2</p> |
| <ul style="list-style-type: none"> · 14.4 Packing group · ADR/RID/ADN, IMDG, IATA | <p style="text-align: center;">III</p> |
| <ul style="list-style-type: none"> · 14.5 Environmental hazards: · Marine pollutant: | <p style="text-align: center;">No</p> |
| <ul style="list-style-type: none"> · 14.6 Special precautions for user: · Hazard identification number (Kemler code): · EMS Number: · Segregation groups · Stowage Category · Handling Code · Segregation Code | <p>Warning: Substances liable to spontaneous combustion.</p> <p style="text-align: center;">40</p> <p style="text-align: center;">F-G,S-J</p> <p>(SGG7) Heavy metals and their salts (including their organometallic compounds), (SGG15) powdered metals</p> <p style="text-align: center;">E</p> <p>H1 Keep as dry as reasonably practicable</p> <p>SG26 In addition: from goods of classes 2.1 and 3 when stowed on deck of a containership a minimum distance of two container spaces athwartship shall be maintained, when stowed on ro-ro ships a distance of 6 m athwartship shall be maintained.</p> |
| <ul style="list-style-type: none"> · 14.7 Maritime transport in bulk according to IMO instruments | <p style="text-align: center;">Not applicable</p> |
| <ul style="list-style-type: none"> · 14.8 Transport/Additional information: | |
| <ul style="list-style-type: none"> · ADR/RID/ADN · Limited quantities (LQ): | <p style="text-align: center;">0</p> |

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· Excepted quantities (EQ)	Code: E1 Maximum net quantity per inner packaging: 30 g Maximum net quantity per outer packaging: 1000 g
· Transport category:	3
· Tunnel restriction code:	E
· IMDG	
· Limited quantities (LQ)	0
· Excepted quantities (EQ)	Code: E1 Maximum net quantity per inner packaging: 30 g Maximum net quantity per outer packaging: 1000 g
· UN "Model Regulation":	UN 3189 METAL POWDER, SELF-HEATING, N.O.S. 4.2, III

SECTION 15: Regulatory information

· 15.1 Safety, health and environmental regulations/legislation specific for the substance or mixture

- Directive 2012/18/EU
- Named dangerous substances - ANNEX I None of the ingredients is listed.
- Seveso category Not applicable
- Qualifying quantity (tonnes) for the application of lower-tier requirements Not applicable
- Qualifying quantity (tonnes) for the application of upper-tier requirements Not applicable

· REGULATION (EU) 2019/1021 on persistent organic pollutants (POP)

None of the ingredients is listed.

· Regulation (EU) No 649/2012

None of the ingredients is listed.

· REGULATION (EU) 2019/1148

· Annex I - RESTRICTED EXPLOSIVES PRECURSORS (Upper limit value for the purpose of licensing under Article 5(3))

None of the ingredients is listed.

· Annex II - REPORTABLE EXPLOSIVES PRECURSORS

None of the ingredients is listed.

· Regulation (EC) No 273/2004 on drug precursors

7723-14-0 Red phosphorus

2A

· Regulation (EC) No 111/2005 laying down rules for the monitoring of trade between the Community and third countries in drug precursors

7723-14-0 Red phosphorus

2

· REGULATION (EC) No 1005/2009 on substances that deplete the ozone layer – ANNEX I (Ozone-depleting potential)

None of the ingredients is listed.

· Other regulations, limitations and prohibitive regulations

· SVHC Candidate List of REACH Regulation Annex XIV Authorisation (14/6/2023)

None of the ingredients is listed

· REACH Regulation Annex XVII Restriction (8/6/2023)

See Section 16 for information about restriction of use.

None of the ingredients is listed

· REACH Regulation Annex XIV Authorisation List (8/4/2022)

None of the ingredients is listed

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· **15.2 Chemical safety assessment:** A Chemical Safety Assessment has not been carried out.

SECTION 16: Other information

· **Recommended restriction of use** Not applicable· **Relevant hazard statements**

H228 Flammable solid.

H252 Self-heating in large quantities; may catch fire.

H270 May cause or intensify fire; oxidiser.

H280 Contains gas under pressure; may explode if heated.

H315 Causes skin irritation.

H412 Harmful to aquatic life with long lasting effects.

· **Classification according to Regulation (EC) No. 1272/2008**

Flammable solids

Self-heating substances and mixtures

Bridging principles

· *****

The contents and format of this SDS are in accordance with Regulation (EC) No 1907/2006, 1272/2008 and Regulation (EU) No 2020/878.

DISCLAIMER OF LIABILITY

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· **Abbreviations and acronyms:**

ADR: Accord relatif au transport international des marchandises dangereuses par route (European Agreement Concerning the International Carriage of Dangerous Goods by Road)

IMDG: International Maritime Code for Dangerous Goods

IATA: International Air Transport Association

GHS: Globally Harmonised System of Classification and Labelling of Chemicals

EINECS: European Inventory of Existing Commercial Chemical Substances

ELINCS: European List of Notified Chemical Substances

CAS: Chemical Abstracts Service (division of the American Chemical Society)

DNEL: Derived No-Effect Level (REACH)

PNEC: Predicted No-Effect Concentration (REACH)

LC50: Lethal concentration, 50 percent

LD50: Lethal dose, 50 percent

PBT: Persistent, Bioaccumulative and Toxic

vPvB: very Persistent and very Bioaccumulative

Ox. Gas 1: Oxidizing gases – Category 1

Press. Gas (Comp.): Gases under pressure – Compressed gas

Flam. Sol. 1: Flammable solids – Category 1

Flam. Sol. 2: Flammable solids – Category 2

Self-heat. 2: Self-heating substances and mixtures – Category 2

Skin Irrit. 2: Skin corrosion/irritation – Category 2

Aquatic Chronic 3: Hazardous to the aquatic environment - long-term aquatic hazard – Category 3

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